DRScaffold: Boosting Dense-Scene Reasoning in Lightweight Vision Language Models

Supplementary Material

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Table of Contents

This supplementary contains additional qualitative examples not included in the main paper appendix. The main appendix retains one representative example per category; all remaining examples appear here.

1. **Additional Accepted Examples** 2
   (a) HyperSim Indoor Scenes 2
   (b) Cityscapes Outdoor Scenes 4
2. **Additional Qualitative Examples on DRBench** 5
   (a) Perception
   (b) Spatial Reasoning
   (c) Affordance Reasoning
   (d) Anomaly Detection
   (e) False Premise Rejection
3. **Additional Qualitative Examples on General Benchmarks** 15
   (a) RealWorldQA
   (b) MMStar
   (c) MMMU
   (d) HallusionBench

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40th Conference on Neural Information Processing Systems (NeurIPS 2026).

A Additional Accepted Examples

Each accepted example shows the source image alongside the question, answer, and structured reasoning fields. One representative example per category appears in the main paper appendix; all additional examples for each category are shown here.

A.1 HyperSim Indoor Scenes

Scene: Bathroom (`ai_001_001`)

Figure 1: HyperSim `ai_001_001`: rendered bathroom with bathtub, sink, shelf, and towel ring.

Accepted | Perception | single-choice

Q: Which object is placed on the shelf above the bathtub and to the right of the yellow rubber duck?
A. Small blue bottle B. Clear glass container C. White soap bar D. Green sponge
Answer: B
Key objects: shelf, rubber duck, glass container
Scene graph: shelf → above → bathtub; rubber duck → on → shelf; glass container → right_of → rubber duck
Reasoning: `[Perception]` Shelf above bathtub holds yellow duck and transparent glass container. `[Relation]` Glass container is to the right of the duck on the same shelf. `[Logic]` Only glass container satisfies both "on shelf" and "right of duck"; all other options absent. `[Conclusion]` Answer is B (clear glass container).

Accepted | False Premise Rejection | open-ended

Q: What color is the showerhead installed on the wall behind the bathtub?
Answer: Not present
Key objects: bathtub, wall, showerhead
Scene graph: bathtub → against → wall, wall → behind → bathtub, showerhead → not_visible → wall
Reasoning: `[Perception]` Wall behind bathtub is tiled and smooth; no fixtures visible. `[Relation]` No showerhead is attached to wall or ceiling above the tub. `[Logic]` If a showerhead were present it would be visible; none is seen. `[Conclusion]` No showerhead exists – premise rejected.

Scene: Bedroom (`ai_001_006`)

Figure 2: HyperSim `ai_001_006`: bedroom with bed, nightstand, dresser, and window.

Accepted | Perception | open-ended

Q: What color is the blanket draped over the bed in front of the window with horizontal blinds?
Answer: Red
Key objects: blanket, bed, window, blinds
Scene graph: blanket → on → bed; bed → in_front_of → window; window → has → horizontal blinds
Reasoning: `[Perception]` Bed with draped blanket located before the window with horizontal blinds. `[Relation]` Blanket covers the bed; bed is positioned in front of the windowed wall. `[Logic]` Only one bed-blanket combination matches the window-anchor; colour is distinctly red. `[Conclusion]` Blanket color is red.

Accepted | Affordance Reasoning | multi-select

Q: Which surfaces can be used to place personal items such as a phone or glasses?
A. Nightstand B. Dresser C. Bed D. Window sill
Answer: A, B
Key objects: nightstand, dresser, bed, window sill
Scene graph: nightstand → left_of → bed, dresser → right_of → bed, bed → supports → pillows, window sill → below → window
Reasoning: `[Perception]` Room contains nightstand and dresser with flat horizontal surfaces, a bed, and a window sill. `[Relation]` Nightstand is beside bed; dresser is against wall; bed surface is soft; sill is narrow and unstable. `[Logic]` Stable, flat, accessible surfaces suited for small items: nightstand (A) and dresser (B). Bed (C) is soft and sloped; window sill (D) is narrow with risk of items falling. `[Conclusion]` Correct selections: A, B.

Scene: Living Room (`ai_001_008`)

Figure 3: HyperSim `ai_001_008`: living room with sofa, armchairs, coffee table, and television.

Accepted | Spatial Reasoning | open-ended

Q: If a person walks from the right armchair toward the TV, what object will they pass directly in front of them first?
Answer: Coffee table
Key objects: armchair (right), coffee table, TV
Scene graph: armchair → right_of → sofa; coffee table → in_front_of → TV; coffee table → between → armchair → and → TV
Reasoning: `[Perception]` Right armchair, coffee table, and TV are aligned along the room's central axis. `[Relation]` Coffee table sits between the armchair and the TV set. `[Logic]` Walking from armchair toward TV, the first object in the path is the coffee table. `[Conclusion]` First obstacle encountered: coffee table.

A.2 Cityscapes Outdoor Scenes

Scene: Berlin street `berlin_000272`

Figure 4: Cityscapes `berlin_000272_000019`: European urban street with parked vehicles and pedestrians.

Accepted | Anomaly Detection | open-ended

Q: Is there any object obstructing the driving lane in front of the silver hatchback on the right?
Answer: Yes
Key objects: silver hatchback, driving lane, bicycle
Scene graph: silver hatchback → on → right_side_of_road, bicycle → near → silver hatchback, bicycle → partially_in → driving_lane
Reasoning: `[Perception]` A silver hatchback is parked on the right side of the road. `[Relation]` A bicycle is parked next to the silver hatchback, extending into the driving lane. `[Logic]` The bicycle's position is really close to the silver hatchback, creating an obstruction. `[Conclusion]` Obstruction present: Yes.

Scene: Berlin street `berlin_000274` **(road works)**

Figure 5: Cityscapes `berlin_000274_000019`: road with orange traffic barriers, yellow plastic bags, and construction signs.

Accepted | Spatial Reasoning | open-ended

Q: If a pedestrian steps onto the sidewalk to the right of the traffic barrier, which object will he/she be standing next to?
Answer: Traffic sign pole
Key objects: sidewalk, traffic barrier, traffic sign pole
Scene graph: sidewalk → right_of → orange traffic barrier, traffic sign pole → on → sidewalk, traffic sign pole → near → traffic barrier
Reasoning: `[Perception]` A traffic sign pole stands on the sidewalk to the right of the barrier. `[Relation]` The sidewalk extends from the road edge, and the traffic sign pole is anchored on it. `[Logic]` If a pedestrian steps onto this sidewalk segment, they would be adjacent to the nearest fixed object—namely, the traffic sign pole. `[Conclusion]` The pedestrian would be standing next to the traffic sign pole.

B Additional Qualitative Examples on DRBench

The following examples compare base model predictions against DRScaffold-trained counterparts on the DRBench test set. Green ✓ = correct; red × = incorrect.

Perception

Cityscapes: `bielefeld_0000 00_039596_leftImg8bit`

[Perception]

Q: What color is the awning attached to the building on the left side of the street, directly above the window with a visible poster?
GT: blue and white striped

InternVL2.5-2B: *"<gray>"* ×
InternVL2.5-2B+DRS: *"blue and white striped"* ✓
An awning is visible on the left side of the building, above a window; it is striped in blue and white colors; the window has a visible poster.

Cityscapes: `bielefeld_0000 00_043389_leftImg8bit`

[Perception]

Q: What color is the vehicle parked directly behind the blue car on the left side of the street?
A. Black B. Silver C. White D. Gray
GT: A

InternVL2.5-2B: *"C"* ×
InternVL2.5-2B+DRS: *"A"* ✓
The blue car is parked on the left side of the street, and the vehicle directly behind it is a dark-colored car. The color of this dark-colored car is black,...

Cityscapes: `bielefeld_000000_048864_leftImg8bit`

[Perception]

Q: What color is the building on the right side of the street that has a dark brick exterior and a blue directional sign attached to it?
A. Gray B. Dark brown C. Light yellow D. White
GT: B

Qwen2.5-VL-3B: *"A. Gray"* ×
Qwen2.5-VL-3B+DRS: *"B"* ✓
B The building on the right side of the street has a dark brown exterior and a blue directional sign attached to it.

Cityscapes: `bielefeld_000000_050426_leftImg8bit`

[Perception]

Q: What color is the car parked directly behind the black sedan with a visible front grille and side mirror?
A. Silver B. Black C. Gray D. White
GT: B

Qwen2.5-VL-3B: *"A. Silver"* ×
Qwen2.5-VL-3B+DRS: *"B"* ✓

Cityscapes: `bielefeld_000000_060786_leftImg8bit`

[Perception]

Q: What color is the traffic light pole located directly behind the woman wearing a striped dress and carrying a brown bag?
GT: gray

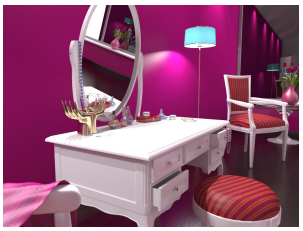
Qwen2.5-VL-3B: *"Blue"* ×
Qwen2.5-VL-3B+DRS: *"Gray"* ✓
Gray ; The pole is dark and consistent with metal construction ; It supports the traffic light and is visible behind the woman ;...

Cityscapes: `mainz_000001_031026_leftImg8bit`

[Perception]

Q: What is the color of the signboard mounted on the construction site fence that displays the company name 'HORN', located to the left of the white banner with 'Fischer-Co'?
A. Blue B. Orange C. White D. Red
GT: B

InternVL2.5-2B: *"C"* ×
InternVL2.5-2B+DRS: *"B"* ✓

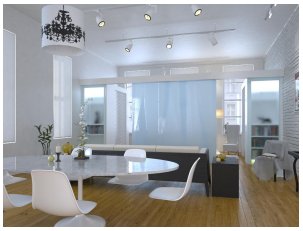


HyperSim: `ai_006_007`

[Perception]

Q: Which objects are visible on the surface of the vanity table? (multiple answers allowed)
A. Gold hand-shaped ornament B. Red jewelry box C. Blue perfume bottle D. Pink vase
GT: A,B,C

InternVL2.5-2B: *"A, B"* ×
InternVL2.5-2B+DRS: *"A, B, C"* ✓
A gold hand-shaped ornament, a red jewelry box, and a blue perfume bottle are clearly visible on the vanity table's surface....

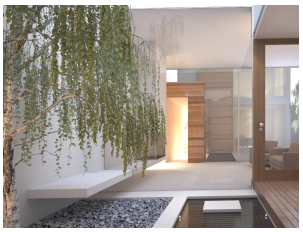


HyperSim: `ai_017_007`

[Perception]

Q: Which object is placed on the round dining table to the left of the central vase?
A. A small green plant B. A yellow candle holder C. A white chair D. A black sculpture
GT: B

Qwen2.5-VL-3B: *"A. A small green plant"* ×
Qwen2.5-VL-3B+DRS: *"B"* ✓

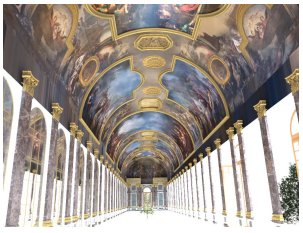


HyperSim: `ai_019_007`

[Perception]

Q: What material is the surface of the white bench made of?
GT: Concrete

Qwen2.5-VL-3B: *"Wood"* ×
Qwen2.5-VL-3B+DRS: *"Concrete"* ✓
Concrete; the bench is smooth and reflective, consistent with concrete.



HyperSim: `ai_023_008`

[Perception]

Q: What shape is the central archway at the far end of the hallway?
GT: Arch

Qwen2.5-VL-3B: *"Octagonal"* ×
Qwen2.5-VL-3B+DRS: *"Arched"* ✓
Arched, with a semi-circular opening.

HyperSim: `ai_038_006`

[Perception]

Q: What material appears to be used for the ceiling panels in the section above the first row of chairs?
GT: Wood

InternVL2.5-2B: *"<metal>"* ×
InternVL2.5-2B+DRS: *"Wood paneling"* ✓
The ceiling above the first row of chairs has a warm, brown tone and grid-like structure; this matches wood paneling.

HyperSim: `ai_047_009`

[Perception]

Q: What material is the floor beneath the conference table?
GT: Carpet

InternVL2.5-2B: *"<wood floor>"* ×
InternVL2.5-2B+DRS: *"<carpeted floor>"* ✓
The floor is visible beneath the conference table and has a textured, patterned appearance consistent with carpeting; no wood or concrete is visible;...

Cityscapes: `bielefeld_000000_046212_leftImg8bit`

[Perception]

Q: What is the color of the signboard mounted above the entrance of the shop on the right side of the street?
A. Blue B. Black C. Red D. Yellow
GT: B

Phi-4-multimodal: *"C"* ×
Phi-4+DRS: *"B"* ✓

HyperSim: `ai_004_004`

[Perception]

Q: What material appears to cover the floor in the living area?
GT: Tile

Phi-4-multimodal: *"Carpet"* ×
Phi-4+DRS: *"tile"* ✓
the floor looks like it has small square tiles on it; the tiles are covering the floor; the tiles look like they could be ceramic or porcelain; the floor is covered by tile.

HyperSim: `ai_001_004`

[Perception]

Q: Which of the following items are visible on the top shelf of the cabinet to the right of the glass-fronted section?
A. Books B. A framed picture C. A vase D. A clock
GT: A

Phi-4-multimodal: *"A,B,C"* ×
Phi-4+DRS: *"A"* ✓

Cityscapes: `mainz_000003_011352_leftImg8bit`

[Perception]

Q: What is the color of the car door that is open on the right side of the street?
A. Black B. Silver C. Gray D. Blue
GT: A

Phi-4-multimodal: *"C"* ×
Phi-4+DRS: *"A"* ✓

Cityscapes: `mainz_000001_034394_leftImg8bit`

[Perception]

Q: What is the color of the compact hatchback driving away from the camera on the right side of the road?
GT: orange

Phi-4-multimodal: *"red"* ×
Phi-4+DRS: *"orange"* ✓
there is an orange car on the right side of the street; it is moving away from the viewer; this is the only vehicle that matches both criteria; the compact hatchback is orange.

Spatial Reasoning

Cityscapes: `bielefeld_000000_050021_leftImg8bit`

[Spatial Reasoning]

Q: What is the position of the woman wearing a pink shirt relative to the group seated at the outdoor tables?
GT: to the left

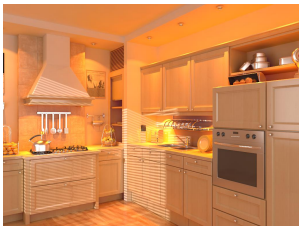
Qwen2.5-VL-3B: *"10"* ×
Qwen2.5-VL-3B+DRS: *"Left of the group"* ✓
Left of the group ; in front of the awning ; not seated

Cityscapes: `bielefeld_000000_059842_leftImg8bit`

[Spatial Reasoning]

Q: If a cyclist were to ride straight from the left sidewalk into the crosswalk, which object would they pass closest to: the blue circular sign or the traffic light pole?
GT: blue circular sign

InternVL2.5-2B: *"the traffic light pole"* ×
InternVL2.5-2B+DRS: *"blue circular sign"* ✓
A cyclist riding straight from the left sidewalk would pass the blue circular sign closest to the crosswalk before reaching the traffic light pole.

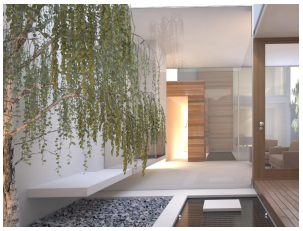


HyperSim: `ai_002_004`

[Spatial Reasoning]

Q: Where will a dropped plate land if it falls from the dish rack to the left of the oven?
A. On the sink B. On the countertop C. On the floor D. In the dishwasher
GT: A

InternVL2.5-2B: *"B"* ×
InternVL2.5-2B+DRS: *"A"* ✓
The plate is on the dish rack, which is above the countertop, and falls downward toward the sink area,...

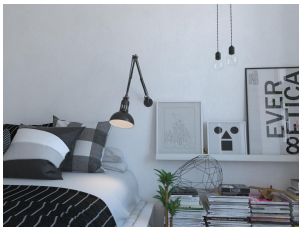


HyperSim: `ai_019_007`

[Spatial Reasoning]

Q: Where would a person standing on the wooden deck be relative to the tree?
A. To the left of the tree B. To the right of the tree C. Behind the tree D. In front of the tree
GT: B

Qwen2.5-VL-3B: *"A. To the left of the tree"* ×
Qwen2.5-VL-3B+DRS: *"B"* ✓



HyperSim: `ai_045_004`

[Spatial Reasoning]

Q: What is the orientation of the plant relative to the stack of books?
GT: Left

InternVL2.5-2B: *"<To the right of the stack of books>"* ×
InternVL2.5-2B+DRS: *"The plant is to the left of the stack of books."* ✓
A green plant is visible to the left of a pile of books on the right side of the image. The books are stacked horizontally, and the plant is positioned in front of them,...

Cityscapes: `mainz_000003_015649_leftImg8bit`

[Spatial Reasoning]

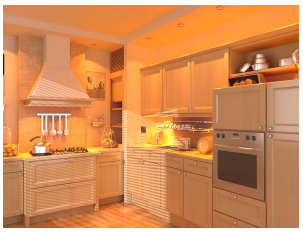
Q: What is the position of the overpass relative to the yellow car and the road below?
GT: above

Phi-4-multimodal: *"ahead"* ×

Phi-4+DRS: *"above"* ✓

The overpass is positioned above the yellow car and the road below. This is evident from the fact that the overpass spans across the entire width of the street,...

Affordance Reasoning



HyperSim: `ai_002_004`

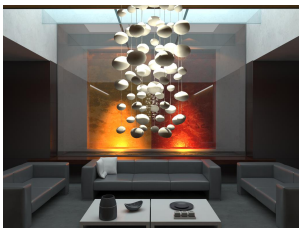
[Functional / Affordance Reasoning]

Q: What is the primary function of the rack mounted above the sink?
GT: Drying dishes

InternVL2.5-2B: *"<Storage and organization of kitchen items>"* ×

InternVL2.5-2B+DRS: *"<Hang utensils and plates for drying>"* ✓

A metal rack is mounted above the sink, holding various utensils and plates; it is positioned to dry after washing....



HyperSim: `ai_005_009`

[Functional / Affordance Reasoning]

Q: Which furniture pieces are suitable for seating multiple people? (multiple answers allowed)
A. Central sofa B. Left armchair C. Right armchair D. Coffee table
GT: A,B,C

InternVL2.5-2B: *"A, C"* ×

InternVL2.5-2B+DRS: *"A, B, C"* ✓

The central sofa and left armchair are designed to accommodate multiple people, with ample seating space and armrests. The right armchair is smaller but still suitable for seating....

HyperSim: `ai_007_007`

[Functional / Affordance Reasoning]

Q: Which objects can be used for sitting at the dining area? (multiple answers allowed)
A. Chair B. Stool C. Bench D. Table
GT: A,B

InternVL2.5-2B: *"A, D"* ×
InternVL2.5-2B+DRS: *"A, B"* ✓
Chairs and stools are visible in the dining area, with chairs having backrests and stools having seats and legs. The table is not designed for sitting but for dining.

HyperSim: `ai_018_004`

[Functional / Affordance Reasoning]

Q: Which of the following surfaces are suitable for placing a drink? (multiple answers allowed)
A. Coffee table B. Kitchen counter C. Bar stool seat D. Sofa armrest
GT: A,B

Qwen2.5-VL-3B: *"A,B,C"* ×
Qwen2.5-VL-3B+DRS: *"A,B"* ✓

Cityscapes: `bonn_000026_000019_leftImg8bit`

[Functional / Affordance Reasoning]

Q: Which surface provides the most stable base for a person to stand on while waiting for the bus at the stop?
GT: sidewalk

Phi-4-multimodal: *"<a short, precise text answer (1-10 words, just the value)>"* ×
Phi-4+DRS: *"sidewalk"* ✓
there is a sidewalk next to the road; sidewalks provide stability; therefore, it would be the most stable surface; sidewalk

Anomaly Detection

Cityscapes: `bielefeld_000000_039082_leftImg8bit`

[Anomaly Detection]

Q: Is there any object obstructing the driving lane in front of the vehicle from which this image is taken?
GT: no

Qwen2.5-VL-3B: *"1"* ×
Qwen2.5-VL-3B+DRS: *"No obstruction is present in the driving lane."* ✓

Cityscapes: `bielefeld_000000_064910_leftImg8bit`

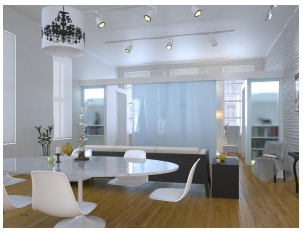
[Anomaly Detection]

Q: Are there any visible signs indicating parking restrictions on the right side of the street?
A. Yes, a blue sign with a red circle B. Yes, a yellow warning sign
C. No signs visible D. Yes, a green parking permit sign
GT: A

InternVL2.5-2B: *"C"* ×

InternVL2.5-2B+DRS: *"A"* ✓

A blue sign with a red circle is visible on the right side of the street, indicating parking restrictions. This matches option A....



HyperSim: `ai_017_007`

[Anomaly Detection]

Q: Is there any object placed in a way that could pose a tripping hazard on the wooden floor?
GT: No

Qwen2.5-VL-3B: *"1"* ×

Qwen2.5-VL-3B+DRS: *"No, there are no loose objects or clutter on the floor."* ✓

Cityscapes: `bielefeld_000000_062121_leftImg8bit`

[Anomaly Detection]

Q: Is there any object obstructing the designated driving path in the center of the road?
GT: yes

Phi-4-multimodal: *"<a short, precise text answer (1-10 words, just the value)>"* ×

Phi-4+DRS: *"<yes>"* ✓

There is an obstacle blocking the lane where vehicles should be driving. So the answer is yes.

False Premise Rejection

Cityscapes: `bielefeld_000000_048754_leftImg8bit`

[False Premise Rejection]

Q: What is the color of the spare tire mounted on the back of the silver hatchback?
GT: none

Qwen2.5-VL-3B: *"The color of the spare tire is black."* ×

Qwen2.5-VL-3B+DRS: *"The spare tire is not visible."* ✓

Cityscapes: `bielefeld_000000_056310_leftImg8bit`

[False Premise Rejection]

Q: What type of emergency equipment is installed on the back of the silver sedan?
GT: none

Qwen2.5-VL-3B: *"1. Fire extinguisher"* ×
Qwen2.5-VL-3B+DRS: *"No emergency equipment visible"* ✓

HyperSim: `ai_001_004`

[False Premise Rejection]

Q: What color is the ceiling fan hanging from the ceiling?
A. White B. Bronze C. Black D. None
GT: D

InternVL2.5-2B: *"A"* ×
InternVL2.5-2B+DRS: *"D"* ✓
no ceiling fan is visible in the image; no fan is present in the scene; the absence of a fan is confirmed by the absence of any fan structure or attachment in the image;...

HyperSim: `ai_008_007`

[False Premise Rejection]

Q: What is the brand name on the monitor screen?
GT: None

Qwen2.5-VL-3B: *"The brand name on the monitor screen is Dell."* ×
Qwen2.5-VL-3B+DRS: *"There is no brand name visible on the monitor screen."* ✓

HyperSim: `ai_021_002`

[False Premise Rejection]

Q: What color is the carpet on the stairs?
A. Gray B. Red C. Blue D. None
GT: D

InternVL2.5-2B: *"A"* ×
InternVL2.5-2B+DRS: *"D"* ✓
no carpet is visible on the stairs; stairs are bare; no fabric or carpet is present; >

Cityscapes: `bielefeld_000000_064805_leftImg8bit`

[False Premise Rejection]

Q: What color is the luggage rack mounted on the roof of the dark-colored station wagon parked on the left side of the street?
A. Black B. Silver C. Red D. None
GT: D

Phi-4-multimodal: *"A"* ×
Phi-4+DRS: *"D"* ✓

C Additional Qualitative Examples on General Benchmarks

The following examples are drawn from four standard VLM benchmarks. In each case the base model answers incorrectly while the DRS fine-tuned counterpart answers correctly, demonstrating that structured spatial CoT training transfers to general visual reasoning.

RealWorldQA

RealWorldQA#0

[RealWorldQA]

Q: In which direction is the front wheel of the car on the right side facing?
A. Left B. Straight C. Right
GT: C

InternVL2.5-2B: *"A"* ×
InternVL2.5-2B+DRS: *"C"* ✓

RealWorldQA#49

[RealWorldQA]

Q: What color is the nearest set of traffic lights?
A. Red B. Green C. Yellow
GT: A

Phi-4-multimodal: *"C"* ×
Phi-4+DRS: *"A"* ✓

RealWorldQA#29

[RealWorldQA]

Q: In which direction is the one-way sign in this scene facing?
A. Left B. Right
GT: B

Qwen2.5-VL-3B: *"A. Left"* ×
Qwen2.5-VL-3B+DRS: *"B. Right"* ✓

RealWorldQA#1

[RealWorldQA]

Q: How many bars are here?
A. There are two parallel bars in this photo. B. There are three bars in the photo. two are the same height. C. There are two bars in this photo. one is high and one is low.
GT: B

InternVL2.5-2B: *"A"* ×
InternVL2.5-2B+DRS: *"B"* ✓

RealWorldQA#60

[RealWorldQA]

Q: Which way does this door open?
A. The door opens outward, swinging to the left. B. The door opens inward, swinging to the left. C. The door opens inward, swinging to the right.
GT: C

Phi-4-multimodal: *"B"* ×
Phi-4+DRS: *"C"* ✓

RealWorldQA#34

[RealWorldQA]

Q: How many vehicles are stopping in the lane to our left?
A. 2 B. 3 C. 4
GT: B

Qwen2.5-VL-3B: *"A. 2"* ×
Qwen2.5-VL-3B+DRS: *"B. 3"* ✓

RealWorldQA#10

[RealWorldQA]

Q: Where is the water compared to the tater tots?
A. The water is behind to the right of the tater tots. B. The water is in front to the right of the tater tots. C. The water is behind to the left of the tater tots.
GT: A

InternVL2.5-2B: *"C"* ×
InternVL2.5-2B+DRS: *"A"* ✓

RealWorldQA#62

[RealWorldQA]

Q: Is the care closest to us facing left, right or straight?
A. Left B. Right C. Straight
GT: A

Phi-4-multimodal: *"C"* ×
Phi-4+DRS: *"A"* ✓

RealWorldQA#80

[RealWorldQA]

Q: What color is the nearest set of traffic lights?
A. Red B. Green C. Yellow
GT: A

Qwen2.5-VL-3B: *"B. Green"* ×
Qwen2.5-VL-3B+DRS: *"A. Red"* ✓

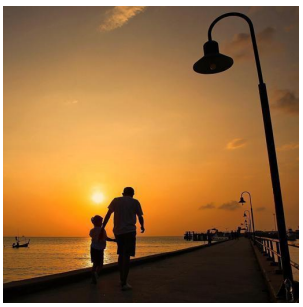
MMStar

MMStar#28
coarse perception/image scene and topic

[MMStar]

Q: What is the primary activity depicted in the image?
A. A crane lifting a box from a building. B. A man on a truck operating a crane to lift a box from a building. C. A crane lifting a house at sunset. D. A man on a balcony of a house under construction in the city.
GT: A

InternVL2.5-2B: *"D"* ×
InternVL2.5-2B+DRS: *"A"* ✓



MMStar#13
coarse perception/image emotion

[MMStar]

Q: Which mood does this image convey?
A. Sad B. Cozy C. Happy D. Angry
GT: B

Phi-4-multimodal: *"C"* ×
Phi-4+DRS: *"B"* ✓

MMStar#12
coarse perception/image scene and topic

[MMStar]

Q: What is the main focus of this image?
A. The grassy hill B. The sculpture C. The fence D. The building
GT: A

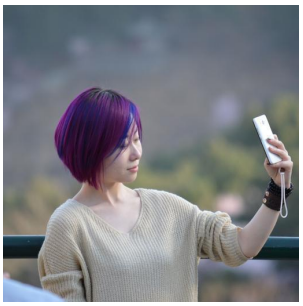
Qwen2.5-VL-3B: *"C. The fence"* ×
Qwen2.5-VL-3B+DRS: *"A. The grassy hill"* ✓

MMStar#52
coarse perception/image scene and topic

[MMStar]

Q: What is the main focus of the image?
A. A fountain with a horse and dragon statue B. A building with sculptures in the front yard C. A park with a statue of a man on a horse D. A statue of a lion and a dragon in front of a building
GT: C

InternVL2.5-2B: *"D"* ×
InternVL2.5-2B+DRS: *"C"* ✓



MMStar#26
coarse perception/image scene and topic

[MMStar]

Q: What is the main focus of the image?
A. the woman's phone B. the woman's hairstyle C. the woman's sweater D. the background scenery
GT: B

Phi-4-multimodal: *"A"* ×
Phi-4+DRS: *"B"* ✓

MMStar#35
coarse perception/image scene and topic

[MMStar]

Q: What is the dominant color in the image?
A. White B. Blue C. Brown D. Black
GT: A

Qwen2.5-VL-3B: *"B. Blue"* ×
Qwen2.5-VL-3B+DRS: *"A. White"* ✓

MMStar#82
coarse perception/image scene and topic

[MMStar]

Q: What is the overall theme of the image?
A. Fashion B. Wildlife C. Nature D. Photography
GT: D

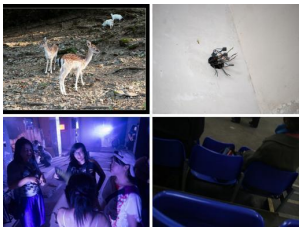
InternVL2.5-2B: *"A"* ×
InternVL2.5-2B+DRS: *"D"* ✓

MMStar#60
coarse perception/image scene and topic

[MMStar]

Q: What is the main color scheme of this image?
A. Black and silver B. White and blue C. Brown and red D. Red and black
GT: D

Phi-4-multimodal: *"C"* ×
Phi-4+DRS: *"D"* ✓



MMStar#42
coarse perception/image style & quality

[MMStar]

Q: Which image shows the highest contrast?
A. upper left B. upper right C. down left D. down right
GT: A

Qwen2.5-VL-3B: *"B. upper right"* ×
Qwen2.5-VL-3B+DRS: *"A. upper left"* ✓

MMMU

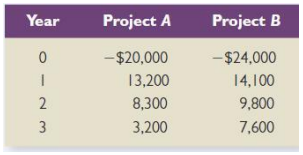
Hudson Alexander, Capital Dec. 1	$175,300
Dec. revenue	56,400
Dec. expenses	59,800

MMMU#2
Business

[MMMU]

Q: Here are facts for the Hudson Roofing Company for December. <image 1> Assuming no investments or withdrawals, what is the ending balance in the owners' capital account?
A. $171,900 B. $170,000 C. $172,500 D. $181,900
GT: A

Phi-4-multimodal: *"D"* ×
Phi-4+DRS: *"A"* ✓



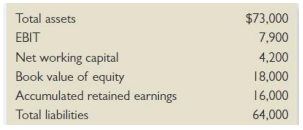
Year	Project A	Project B
0	−$20,000	−$24,000
1	13,200	14,100
2	8,300	9,800
3	3,200	7,600

MMMU#388
Business

[MMMU]

Q: Maxwell Software, Inc., has the following mutually exclusive projects.Suppose the company uses the NPV rule to rank these two projects.<image 1> Which project should be chosen if the appropriate discount rate is 15 percent?
A. Project A B. Project B
GT: B

Phi-4-multimodal: *"A. Project A"* ×
Phi-4+DRS: *"B"* ✓



Total assets	$73,000
EBIT	7,900
Net working capital	4,200
Book value of equity	18,000
Accumulated retained earnings	16,000
Total liabilities	64,000

MMMU#407
Business

[MMMU]

Q: Jon Fulkerson has received a credit application from Seether, LLC, a private company. An abbreviated portion of the financial information provided by the company is shown below:<image 1>What is the Z-score for this company?
A. 3.470 B. 3.096 C. 3.187
GT: B

Qwen2.5-VL-3B: *"To calculate the Z-score for Seether, LLC, we need to use the following formula: \[. . . "* ×
Qwen2.5-VL-3B+DRS: *"B. 3.096"* ✓

MMMU#723
Tech & Engineering

[MMMU]

Q: <image 1> What is the common term for the yellow area surrounding the site of an infection?
A. I don't know and I don't want to guess B. Corona C. Border D. Halo
GT: D

InternVL2.5-2B: *"C"* ×
InternVL2.5-2B+DRS: *"D"* ✓

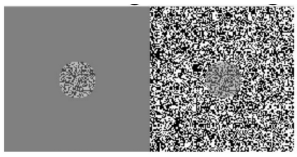
MMMU#708
Tech & Engineering

[MMMU]

Q: <image 1> What causes these unusual formations on Mountain papaya?
A. Abiotic B. Confused C. Biotic D. Normal
GT: C

Qwen2.5-VL-3B: *"A. Abiotic"* ×
Qwen2.5-VL-3B+DRS: *"C. Biotic"* ✓

HallusionBench



`HallusionBench#VD_illusion_1_5_0_2`
VD_illusion

[HallusionBench]

Q: The image shows the famous Chubb illusion. Is the center area on the left smaller than the center area on the right?
GT: No

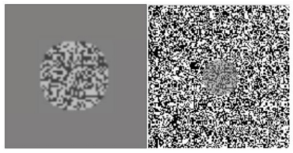
InternVL2.5-2B: *"Yes"* ×
InternVL2.5-2B+DRS: *"No"* ✓

`HallusionBench#VD_illusion_1_1_0_1`
VD_illusion

[HallusionBench]

Q: Is the right grey box smaller than the left grey box?
GT: No

Qwen2.5-VL-3B: *"The grey box on the right is smaller than the grey box on the left."* ×
Qwen2.5-VL-3B+DRS: *"No, the right grey box is not smaller than the left grey box. "* ✓

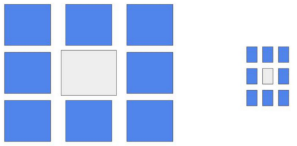


`HallusionBench#VD_illusion_2_5_1_2`
VD_illusion

[HallusionBench]

Q: The image shows the famous Chubb illusion. Is the center area on the left smaller than the center area on the right?
GT: No

InternVL2.5-2B: *"Yes"* ×
InternVL2.5-2B+DRS: *"no"* ✓



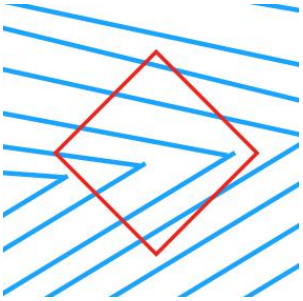
HallusionBench#VD_illusion_2_1_1_0
VD_illusion

[HallusionBench]

Q: Is the right grey box larger than the left grey box?
GT: No

Qwen2.5-VL-3B: *"The right grey box is larger than the left grey box."* ×

Qwen2.5-VL-3B+DRS: *"No, the right grey box is not larger than the left grey box."* ✓



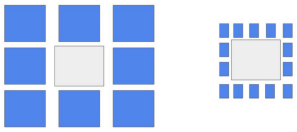
HallusionBench#VD_illusion_1_14_0_1
VD_illusion

[HallusionBench]

Q: This image shows the orbison illusion. Whether the red shape in the image is a square? Yes or No
GT: Yes

InternVL2.5-2B: *"No"* ×

InternVL2.5-2B+DRS: *"Yes"* ✓



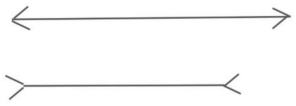
HallusionBench#VD_illusion_1_1_0_1
VD_illusion

[HallusionBench]

Q: Is the right grey box smaller than the left grey box?
GT: No

Phi-4-multimodal: *"The right grey box is smaller than the left grey box. The left grey box is a 3x3 grid of blue square"* ×

Phi-4+DRS: *"No"* ✓



HallusionBench#VD_illusion_2_2_1_0
VD_illusion

[HallusionBench]

Q: Is the top line longer than the bottom line?
GT: Yes

Qwen2.5-VL-3B: *"No, the top line is not longer than the bottom line. Both lines are of equal length."* ×

Qwen2.5-VL-3B+DRS: *"Yes, the top line is longer than the bottom line."* ✓